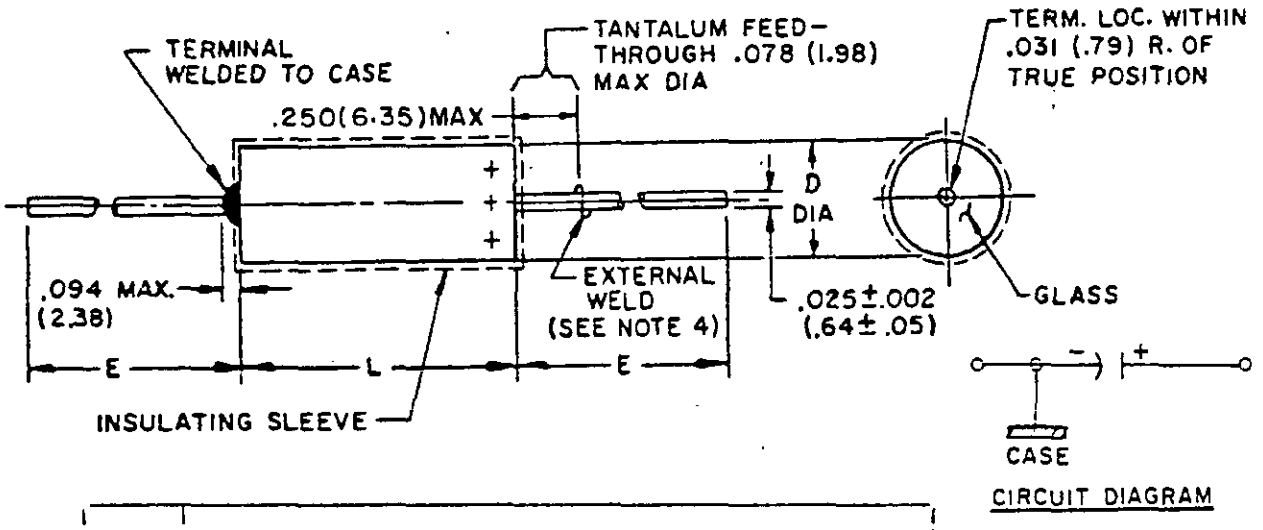
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), 85°C (VOLTAGE DERATED TO 125°C), ESTABLISHED RELIABILITY, STYLE CLR79

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-39006.



Case size	Dimensions			
	Basic case		Insulated Case	
	L 1/ +.031(0.79) -.016(0.41)	D +.016(0.41)	D max	E ±.250(6.35)
T1	.453(11.51)	.188(4.78)	.219 (5.56)	1.500(38.10)
T2	.641(16.28)	.281(7.14)	.312 (7.92)	2.250(57.15)
T3	.766(19.46)	.375(9.52)	.406 (10.31)	2.250(57.15)
T4	1.062(26.97)	.375(9.52)	.406 (10.31)	2.250(57.15)

1/ Length of basic case sleeving shall be as specified in MIL-C-39006.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Style CLR79 capacitors.

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DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

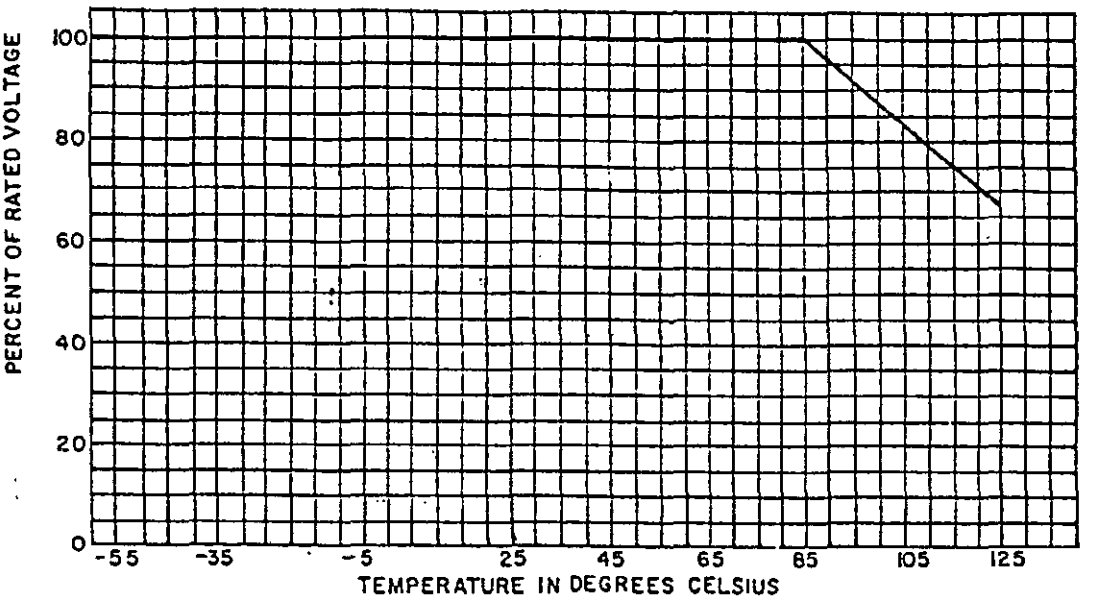


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tantalum, tubular, insulated.

Seal type: Hermetic (glass-to-tantalum anode). For compression type; see table I.

Terminals: Axial-wire lead:

Cathode and anode (type N4 or N5 of MIL-STD-1276).

Rated temperature: -55°C to +85°C, voltage derated to +125°C (see figure 2).

Rated voltage: See table I.

Capacitance (Cap.) nominal (nom.): See table I.

Cap. tolerance: See table I.

Failure rate level: M, P, R, and S (see table I).

DC leakage (DCL) (maximum) (at 25°C): See table I.

Cap. (nom.): Method 305 of MIL-STD-202. See table I.

Dissipation factor (DF) (maximum): See table I.

Seal: Method 112 of MIL-STD-202, conditions A and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100g) or condition D (500g), as applicable.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20g) or condition H (80g), as applicable.

Random vibration (when applicable): Method 214 of MIL-STD-202, test condition II-K (51g).

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Thermal shock: Method 107 of MIL-STD-202, condition A (with step 3 at +125°C).

Number of cycles: 300 for qualification and group C; 30 for group B.

DCL: 200 percent (maximum) of 25°C value of table I for qualification and group C.
DCL: 125 percent (maximum) of 25°C value of table I for gorup B.
ΔCap.: Within ±5 percent of initial measurement.
DF: 115 percent (maximum) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Solderability: Method 208 of MIL-STD-202.

Terminal strength:

Pull: Method 211 of MIL-STD-202, condition A.

Wire-lead bend: In accordance with MIL-C-39006.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.

DCL: 125 percent (maximum) of 25°C value of table I.
ΔCap.: Within ±8 percent of initial measurement.
DF: 115 percent (maximum) of initial requirement.

Dielectric withstanding voltage (DWV):

Method 301 of MIL-STD-202. 2,000 V dc, minimum.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B.
100 megohms, minimum.

Low temperature (storage): Method 502 of MIL-STD-810.

DCL: See table I.
ΔCap.: Within ±5 percent of initial measurement.
DF: See table I.

Stability at low and high temperatures:

Step 1 (+25°C):

DCL: See table I.
Cap.: Within tolerance of table I value.
DF: See table I.

Step 2 (-55°C):

Impedance: See table I.
ΔCap.: See table I.

Step 3 (+25°C):

DCL: See table I.
ΔCap.: Within ±5 percent of step 1 value.
DF: See table I.

Step 4 (+85°C):

DCL: See table I.
ΔCap.: See table I.
DF: See table I.

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Step 5 (+125°C):

DCL: See table I.
ΔCap.: See table I.
DF: See table I.

Step 6 (+25°C):

DCL: See table I.
ΔCap.: Within ±5 percent of step 1 value.
DF: See table I.

Reverse voltage: As specified in MIL-C-39006.

DCL: See table I.
Cap.: Within tolerance of table I value.
DF: See table I.

Life at 85°C: Method 108 of MIL-STD-202.

2,000-hour (qualification):

DCL: See table I.
ΔCap.: Within ±10 percent of initial measurement.
DF: See table I.

10,000-hour (group B) and extended life:

DCL (85°C): 125 percent (maximum) of table I value.
DCL (25°C): See table I.
ΔCap.: Within +10 percent, -20 percent of initial measurement.
DF: 200 percent (maximum) of table I value.
DWV: 2,000 V dc, minimum.
IR: 100 megohms, minimum.

AC ripple life at 85°C (qualification and group C):

Test ripple current: See table I.
Test voltage: 66.67 percent of rated voltage (see table I).
DCL: See table I.
ΔCap.: Within ±10 percent of initial measurement.
DF: See table I.

Vacuum life: Not applicable.

Life at 125°C: Method 108 of MIL-STD-202.

DCL: See table I.
ΔCap.: Within ±10 percent of initial measurement.
DF: See table I.
DWV: 2,000 V dc, minimum.
IR: 100 megohms, minimum.

10,000-hour (group B):

DCL (125°C): 125 percent (maximum) of table I value.
DCL (25°C): See table I.
ΔCap.: Within +10 percent, -20 percent of initial measurement.
DF: 200 percent (maximum) of table I value.
DWV: 2,000 V dc, minimum.
IR: 100 megohms, minimum.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 feet).

Test voltage: Rated voltage.

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Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

DCL: See table I.
ΔCap.: Within ±5 percent of initial measurement.
DF: See table I.

Marking: In accordance with MIL-C-39006, with the following exception: High frequency parts: The manufacturer may mark the T1 case capacitors as follows:

Example: M39006/
22H0001
JAN TM
12345
B415A
30 µF 6 V

Voltage groups (85°C rated voltage):

I - - - - - - - - - - 6 to 15 V inclusive
II- - - - - - - - - - 25 to 75 V inclusive
III - - - - - - - - - 100 and 125 V inclusive

Substitutability data: See table III.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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TABLE I. Capacitor characteristics.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.)		Dissipation factor (max.)	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at			2/ Max 85°C 40 kHz ripple current	Case size	Part no. M39006/22- 3/ noncompression seal				Part no. M39006/22- 3/ compression seal			
			+25°C	+85°C & +125°C					-55°C	+85°C	+125°C			Failure rate level (%/1,000 hr.)				Failure rate level (%/1,000 hr.)			
														M(1.0)	P(0.1)	R(0.01)	S(0.001)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	V dc	V dc	Ohms	%	%	%	mA rms									
6	30	±20	1	2	9	4	6.9	100	-40	+10.5	+12	820	T1	0001-	0221-	0441-	0661-	1001-	2011-	3041-	3061-
"	30	±10	"	"	"	"	"	100	"	"	"	820	"	0002-	0222-	0442-	0662-	1002-	2012-	3042-	3062-
"	30	±5	"	"	"	"	"	100	"	"	"	820	"	0003-	0223-	0443-	0663-	1003-	2013-	3043-	3063-
"	63	±20	"	"	15	"	"	60	"	+14	+16	960	"	0004-	0224-	0444-	0664-	1004-	2014-	3044-	3064-
"	68	±10	"	"	"	"	"	60	"	"	"	960	"	0005-	0225-	0445-	0665-	1005-	2015-	3045-	3065-
"	68	±5	"	"	"	"	"	60	"	"	"	960	"	0006-	0226-	0446-	0666-	1006-	2016-	3046-	3066-
"	140	±20	"	3	21	"	"	40	"	"	"	1200	T2	0007-	0227-	0447-	0667-	1007-	2017-	3047-	3067-
"	140	±10	"	"	"	"	"	40	"	"	"	1200	"	0008-	0228-	0448-	0668-	1008-	2018-	3048-	3068-
"	140	±5	"	"	"	"	"	40	"	"	"	1200	"	0009-	0229-	0449-	0669-	1009-	2019-	3049-	3069-
"	270	±20	"	6.5	45	"	"	25	-44	+17.5	+20	1375	"	0010-	0230-	0450-	0670-	1010-	2020-	3050-	3070-
"	270	±10	"	"	"	"	"	25	"	"	"	1375	"	0011-	0231-	0451-	0671-	1011-	2021-	3051-	3071-
"	270	±5	"	"	"	"	"	25	"	"	"	1375	"	0012-	0232-	0452-	0672-	1012-	2022-	3052-	3072-
"	330	±20	2	7.9	36	"	"	20	"	+14	+16	1800	T3	0013-	0233-	0453-	0673-	1013-	2023-	3053-	3073-
"	330	±10	"	"	"	"	"	20	"	"	"	1800	"	0014-	0234-	0454-	0674-	1014-	2024-	3054-	3074-
"	330	±5	"	"	"	"	"	20	"	"	"	1800	"	0015-	0235-	0455-	0675-	1015-	2025-	3055-	3075-
"	560	±20	"	13	55	"	"	25	-64	+17.5	+20	1900	"	0016-	0236-	0456-	0676-	1016-	2026-	3056-	3076-
"	560	±10	"	"	"	"	"	25	"	"	"	1900	"	0017-	0237-	0457-	0677-	1017-	2027-	3057-	3077-
"	560	±5	"	"	"	"	"	25	"	"	"	1900	"	0018-	0238-	0458-	0678-	1018-	2028-	3058-	3078-
"	1200	±20	3	14	90	"	"	20	-80	+25	+25	2265	T4	0019-	0239-	0459-	0679-	1019-	2029-	3059-	3079-
"	1200	±10	3	14	90	"	"	20	-80	+25	+25	2265	T4	0020-	0240-	0460-	0680-	1020-	2030-	3060-	3080-
8	25	±20	1	2	7.5	5	9.2	100	-40	+10.5	+12	820	T1	0021-	0241-	0461-	0681-	1021-	2031-	3061-	3081-
"	25	±10	"	"	"	"	"	100	"	"	+12	820	"	0022-	0242-	0462-	0682-	1022-	2032-	3062-	3082-
"	25	±5	"	"	"	"	"	100	"	"	+12	820	"	0023-	0243-	0463-	0683-	1023-	2033-	3063-	3083-
"	56	±20	"	"	14	"	"	59	"	+14	+16	900	"	0024-	0244-	0464-	0684-	1024-	2034-	3064-	3084-
"	56	±10	"	"	"	"	"	59	"	"	+16	900	"	0025-	0245-	0465-	0685-	1025-	2035-	3065-	3085-
"	56	±5	"	"	"	"	"	59	"	"	+16	900	"	0026-	0246-	0466-	0686-	1026-	2036-	3066-	3086-
"	120	±20	"	"	20	"	"	50	-44	+17.5	+20	1220	T2	0027-	0247-	0467-	0687-	1027-	2037-	3067-	3087-
"	120	±10	"	"	"	"	"	50	"	"	"	1220	"	0028-	0248-	0468-	0688-	1028-	2038-	3068-	3088-
"	120	±5	"	"	"	"	"	50	"	"	"	1220	"	0029-	0249-	0469-	0689-	1029-	2039-	3069-	3089-
"	220	±20	"	7	37	"	"	30	"	"	"	1370	"	0030-	0250-	0470-	0690-	1030-	2040-	3070-	3090-
"	220	±10	"	"	"	"	"	30	"	"	"	1370	"	0031-	0251-	0471-	0691-	1031-	2041-	3071-	3091-
"	220	±5	"	"	"	"	"	30	"	"	"	1370	"	0032-	0252-	0472-	0692-	1032-	2042-	3072-	3092-
"	290	±20	2	6	34	"	"	25	-64	"	"	1770	T3	0033-	0253-	0473-	0693-	1033-	2043-	3073-	3093-
"	290	±10	"	"	"	"	"	"	"	"	"	1770	"	0034-	0254-	0474-	0694-	1034-	2044-	3074-	3094-
"	290	±5	"	"	"	"	"	"	"	"	"	1770	"	0035-	0255-	0475-	0695-	1035-	2045-	3075-	3095-
"	430	±20	"	14	46	"	"	"	"	"	"	1825	"	0036-	0256-	0476-	0696-	1036-	2046-	3076-	3096-
"	430	±10	"	"	"	"	"	"	"	"	"	1825	"	0037-	0257-	0477-	0697-	1037-	2047-	3077-	3097-
"	430	±5	"	"	"	"	"	"	"	"	"	1825	"	0038-	0258-	0478-	0698-	1038-	2048-	3078-	3098-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.) +25°C	DC leakage (max.) +85°C & +125°C	Dissipation factor (max.)	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	2/ Max 85°C 40 kHz ripple current	Case size	Part no. M39006/22- 3/ noncompression seal — Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)	Part no. M39006/22- 3/ compression seal — Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	uF	%	uA	uA	%	V dc	V dc	Ohms	%	%	%	mA rms									
8	850	±20	4	16	60	5	9.2	22	-80	+25	+25	2330	T4	0039-	0259-	0479-	0699-	1039-	2049-	3079-	3099-
8	850	±10	4	16	60	5	9.2	22	-80	+25	+25	2330	T4	0040-	0260-	0480-	0700-	1040-	2050-	3080-	3100-
10	20	±20	1	2	6	7	11.5	175	-32	+10.5	+12	820	T1	0041-	0261-	0481-	0701-	1041-	2051-	3081-	3101-
"	20	±10	"	"	"	"	"	175	"	"	"	820	"	0042-	0262-	0482-	0702-	1042-	2052-	3082-	3102-
"	20	±5	"	"	"	"	"	175	"	"	"	820	"	0043-	0263-	0483-	0703-	1043-	2053-	3083-	3103-
"	47	±20	"	"	13	"	"	100	-36	+14	+16	855	"	0044-	0264-	0484-	0704-	1044-	2054-	3084-	3104-
"	47	±10	"	"	"	"	"	100	"	"	"	855	"	0045-	0265-	0485-	0705-	1045-	2055-	3085-	3105-
"	47	±5	"	"	"	"	"	100	"	"	"	855	"	0046-	0266-	0486-	0706-	1046-	2056-	3086-	3106-
"	100	±20	"	4	15	"	"	60	"	"	"	1200	T2	0047-	0267-	0487-	0707-	1047-	2057-	3087-	3107-
"	100	±10	"	"	"	"	"	60	"	"	"	1200	"	0048-	0268-	0488-	0708-	1048-	2058-	3088-	3108-
"	100	±5	"	"	"	"	"	60	"	"	"	1200	"	0049-	0269-	0489-	0709-	1049-	2059-	3089-	3109-
"	180	±20	"	7	30	"	"	40	"	"	"	1365	"	0050-	0270-	0490-	0710-	1050-	2060-	3090-	3110-
"	180	±10	"	"	"	"	"	40	"	"	"	1365	"	0051-	0271-	0491-	0711-	1051-	2061-	3091-	3111-
"	180	±5	"	"	"	"	"	40	"	"	"	1365	"	0052-	0272-	0492-	0712-	1052-	2062-	3092-	3112-
"	250	±20	2	10	"	"	"	30	-40	"	"	1720	T3	0053-	0273-	0493-	0713-	1053-	2063-	3093-	3113-
"	250	±10	"	"	"	"	"	30	"	"	"	1720	"	0054-	0274-	0494-	0714-	1054-	2064-	3094-	3114-
"	250	±5	"	"	"	"	"	30	"	"	"	1720	"	0055-	0275-	0495-	0715-	1055-	2065-	3095-	3115-
"	390	±20	"	16	44	"	"	25	-64	+17.5	+20	1800	"	0056-	0276-	0496-	0716-	1056-	2066-	3096-	3116-
"	390	±10	"	"	"	"	"	25	"	"	"	1800	"	0057-	0277-	0497-	0717-	1057-	2067-	3097-	3117-
"	390	±5	"	"	"	"	"	25	"	"	"	1800	"	0058-	0278-	0498-	0718-	1058-	2068-	3098-	3118-
"	750	±20	4	"	50	"	"	23	-80	+25	+25	2360	T4	0059-	0279-	0499-	0719-	1059-	2069-	3099-	3119-
"	750	±10	4	"	"	"	"	23	-80	"	+25	2360	T4	0060-	0280-	0500-	0720-	1060-	2070-	3100-	3120-
15	15	±20	1	2	5	10	17.2	155	-24	+10.5	+12	780	T1	0061-	0281-	0501-	0721-	1061-	2071-	3101-	3121-
"	15	±10	"	"	"	"	"	155	"	"	"	780	"	0062-	0282-	0502-	0722-	1062-	2072-	3102-	3122-
"	15	±5	"	"	"	"	"	155	"	"	"	780	"	0063-	0283-	0503-	0723-	1063-	2073-	3103-	3123-
"	33	±20	"	"	10	"	"	90	-28	+14	+16	820	"	0064-	0284-	0504-	0724-	1064-	2074-	3104-	3124-
"	33	±10	"	"	"	"	"	90	"	"	"	820	"	0065-	0285-	0505-	0725-	1065-	2075-	3105-	3125-
"	33	±5	"	"	"	"	"	90	"	"	"	820	"	0066-	0286-	0506-	0726-	1066-	2076-	3106-	3126-
"	70	±20	"	4	13	"	"	75	"	"	"	1150	T2	0067-	0287-	0507-	0727-	1067-	2077-	3107-	3127-
"	70	±10	"	"	"	"	"	75	"	"	"	1150	"	0068-	0288-	0508-	0728-	1068-	2078-	3108-	3128-
"	70	±5	"	"	"	"	"	75	"	"	"	1150	"	0069-	0289-	0509-	0729-	1069-	2079-	3109-	3129-
"	120	±20	"	7	18	"	"	50	"	+17.5	+20	1450	"	0070-	0290-	0510-	0730-	1070-	2080-	3110-	3130-
"	120	±10	"	"	"	"	"	50	"	"	"	1450	"	0071-	0291-	0511-	0731-	1071-	2081-	3111-	3131-
"	120	±5	"	"	"	"	"	50	"	"	"	1450	"	0072-	0292-	0512-	0732-	1072-	2082-	3112-	3132-
"	170	±20	2	10	25	"	"	35	-32	+14	+16	1480	T3	0073-	0293-	0513-	0733-	1073-	2083-	3113-	3133-
"	170	±10	"	"	"	"	"	35	"	"	"	1480	"	0074-	0294-	0514-	0734-	1074-	2084-	3114-	3134-
"	170	±5	"	"	"	"	"	35	"	"	"	1480	"	0075-	0295-	0515-	0735-	1075-	2085-	3115-	3135-
"	270	±20	"	16	32	"	"	30	-56	+17.5	+20	1740	"	0076-	0296-	0516-	0736-	1076-	2086-	3116-	3136-
"	270	±10	"	"	"	"	"	30	"	"	"	1740	"	0077-	0297-	0517-	0737-	1077-	2087-	3117-	3137-
"	270	±5	"	"	"	"	"	30	"	"	"	1740	"	0078-	0298-	0518-	0738-	1078-	2088-	3118-	3138-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.)		Dissipation factor (max.)	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at			2/ Max 85°C 40 kHz ripple current	Case size	Part no. M39006/22- 3/ noncompression seal				Part no. M39006/22- 3/ compression seal			
			+25°C	+85°C & +125°C					-55°C	+85°C	+125°C			Failure rate level (%/1,000 hr.)				Failure rate level (%/1,000 hr.)			
														M(1.0)	P(0.1)	R(0.01)	S(0.001)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	V dc	V dc	Ohms	%	%	%	mA rms									
15	540	±20	6	24	40	10	17.2	23	-80	+25	+25	2330	T4	0079-	0299-	0519-	0739-	1079-	2089-	3119-	3139-
15	540	±10	6	24	40	10	17.2	23	-80	+25	+25	2330	T4	0080-	0300-	0520-	0740-	1080-	2090-	3120-	3140-
25	101	±20	1	2	4	15	28.8	220	-16	+8	+9	715	T1	0081-	0301-	0521-	0741-	1081-	2091-	3121-	3141-
"	101	±10	"	"	"	"	"	220	"	+8	"	715	"	0082-	0302-	0522-	0742-	1082-	2092-	3122-	3142-
"	101	±5	"	"	"	"	"	220	"	+8	"	715	"	0083-	0303-	0523-	0743-	1083-	2093-	3123-	3143-
"	221	±20	"	"	6.6	"	"	140	-20	+10.5	+12	825	"	0084-	0304-	0524-	0744-	1084-	2094-	3124-	3144-
"	221	±10	"	"	"	"	"	140	"	"	"	825	"	0085-	0305-	0525-	0745-	1085-	2095-	3125-	3145-
"	221	±5	"	"	"	"	"	140	"	"	"	825	"	0086-	0306-	0526-	0746-	1086-	2096-	3126-	3146-
"	501	±20	"	"	11.0	"	"	70	-28	+13	+15	1130	T2	0087-	0307-	0527-	0747-	1087-	2097-	3127-	3147-
"	501	±10	"	"	"	"	"	70	"	"	"	1130	"	0088-	0308-	0528-	0748-	1088-	2098-	3128-	3148-
"	501	±5	"	"	"	"	"	70	"	"	"	1130	"	0089-	0309-	0529-	0749-	1089-	2099-	3129-	3149-
"	1001	±20	"	10	15	"	"	50	"	"	"	1435	"	0090-	0310-	0530-	0750-	1090-	2100-	3130-	3150-
"	1001	±10	"	"	"	"	"	50	"	"	"	1435	"	0091-	0311-	0531-	0751-	1091-	2101-	3131-	3151-
"	1001	±5	"	"	"	"	"	50	"	"	"	1435	"	0092-	0312-	0532-	0752-	1092-	2102-	3132-	3152-
"	1201	±20	2	6	21	"	"	38	-32	"	"	1450	T3	0093-	0313-	0533-	0753-	1093-	2103-	3133-	3153-
"	1201	±10	"	"	"	"	"	38	"	"	"	1450	"	0094-	0314-	0534-	0754-	1094-	2104-	3134-	3154-
"	1201	±5	"	"	"	"	"	38	"	"	"	1450	"	0095-	0315-	0535-	0755-	1095-	2105-	3135-	3155-
"	1801	±20	"	18	26	"	"	32	-48	"	"	1525	"	0096-	0316-	0536-	0756-	1096-	2106-	3136-	3156-
"	1801	±10	"	"	"	"	"	32	"	"	"	1525	"	0097-	0317-	0537-	0757-	1097-	2107-	3137-	3157-
"	1801	±5	"	"	"	"	"	32	"	"	"	1525	"	0098-	0318-	0538-	0758-	1098-	2108-	3138-	3158-
"	3501	±20	7	28	35	"	"	24	-70	+25	+25	1970	T4	0099-	0319-	0539-	0759-	1099-	2109-	3139-	3159-
"	3501	±10	7	28	35	"	"	24	-70	+25	+25	1970	T4	0100-	0320-	0540-	0760-	1100-	2110-	3140-	3160-
30	81	±20	1	2	4	20	34.5	275	-16	+8	+12	640	T1	0101-	0321-	0541-	0761-	1101-	2111-	3141-	3161-
"	81	±10	"	"	"	"	"	275	"	+8	"	640	"	0102-	0322-	0542-	0762-	1102-	2112-	3142-	3162-
"	81	±5	"	"	"	"	"	275	"	+8	"	640	"	0103-	0323-	0543-	0763-	1103-	2113-	3143-	3163-
"	151	±20	"	"	5	"	"	175	-20	+10.5	"	780	"	0104-	0324-	0544-	0764-	1104-	2114-	3144-	3164-
"	151	±10	"	"	"	"	"	175	"	"	"	780	"	0105-	0325-	0545-	0765-	1105-	2115-	3145-	3165-
"	151	±5	"	"	"	"	"	175	"	"	"	780	"	0106-	0326-	0546-	0766-	1106-	2116-	3146-	3166-
"	401	±20	"	5	10	"	"	65	-24	"	"	1120	T2	0107-	0327-	0547-	0767-	1107-	2117-	3147-	3167-
"	401	±10	"	"	"	"	"	65	"	"	"	1120	"	0108-	0328-	0548-	0768-	1108-	2118-	3148-	3168-
"	401	±5	"	"	"	"	"	65	"	"	"	1120	"	0109-	0329-	0549-	0769-	1109-	2119-	3149-	3169-
"	681	±20	"	8	13	"	"	60	"	+13	+15	1285	"	0110-	0330-	0550-	0770-	1110-	2120-	3150-	3170-
"	681	±10	"	"	"	"	"	60	"	"	"	1285	"	0111-	0331-	0551-	0771-	1111-	2121-	3151-	3171-
"	681	±5	"	"	"	"	"	60	"	"	"	1285	"	0112-	0332-	0552-	0772-	1112-	2122-	3152-	3172-
"	1001	±20	2	12	17	"	"	40	-28	+10.5	+12	1450	T3	0113-	0333-	0553-	0773-	1113-	2123-	3153-	3173-
"	1001	±10	"	"	"	"	"	40	"	"	"	1450	"	0114-	0334-	0554-	0774-	1114-	2124-	3154-	3174-
"	1001	±5	"	"	"	"	"	40	"	"	"	1450	"	0115-	0335-	0555-	0775-	1115-	2125-	3155-	3175-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.)		Dissipation factor (max.)	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at			2/ Max 85°C 40 kHz ripple current	Case size	Part no. M39006/22- 3/ noncompression seal				Part no. M39006/22- 3/ compression seal			
			+25°C	+85°C & +125°C					-55°C	+85°C	+125°C			Failure rate level (%/1,000 hr.)				Failure rate level (%/1,000 hr.)			
														M(1.0)	P(0.1)	R(0.01)	S(0.001)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	V dc	V dc	Ohms	%	%	%	mA rms									
30	150	±20	2	18	23	20	34.5	35	-48	+13	+15	1525	T3	0116-	0336-	0556-	0776-	1116-	2126-	3156-	3176-
"	150	±10	"	"	"	"	"	35	-48	+13	+15	1525	"	0117-	0337-	0557-	0777-	1117-	2127-	3157-	3177-
"	150	±5	"	"	"	"	"	35	-48	+13	+15	1525	"	0118-	0338-	0558-	0778-	1118-	2128-	3158-	3178-
"	300	±20	8	32	31	"	"	25	-60	+25	+25	1950	T4	0119-	0339-	0559-	0779-	1119-	2129-	3159-	3179-
"	300	±10	8	32	31	"	"	25	-60	+25	+25	1950	T4	0120-	0340-	0560-	0780-	1120-	2130-	3160-	3180-
50	5	±20	1	2	3	30	57.5	400	-16	+5	+6	580	T1	0121-	0341-	0561-	0781-	1121-	2131-	3161-	3181-
"	5	±10	"	"	"	"	"	400	-16	+5	+6	580	"	0122-	0342-	0562-	0782-	1122-	2132-	3162-	3182-
"	5	±5	"	"	"	"	"	400	-16	+5	+6	580	"	0123-	0343-	0563-	0783-	1123-	2133-	3163-	3183-
"	10	±20	"	"	4	"	"	250	-24	+8	+9	715	"	0124-	0344-	0564-	0784-	1124-	2134-	3164-	3184-
"	10	±10	"	"	"	"	"	250	-24	+8	+9	715	"	0125-	0345-	0565-	0785-	1125-	2135-	3165-	3185-
"	10	±5	"	"	"	"	"	250	-24	+8	+9	715	"	0126-	0346-	0566-	0786-	1126-	2136-	3166-	3186-
"	25	±20	"	5	8	"	"	95	-20	+10.5	+12	1005	T2	0127-	0347-	0567-	0787-	1127-	2137-	3167-	3187-
"	25	±10	"	"	"	"	"	95	-20	+10.5	+12	1005	"	0128-	0348-	0568-	0788-	1128-	2138-	3168-	3188-
"	25	±5	"	"	"	"	"	95	-20	+10.5	+12	1005	"	0129-	0349-	0569-	0789-	1129-	2139-	3169-	3189-
"	47	±20	"	9	11	"	"	70	-28	+13	+15	1155	"	0130-	0350-	0570-	0790-	1130-	2140-	3170-	3190-
"	47	±10	"	"	"	"	"	70	-28	+13	+15	1155	"	0131-	0351-	0571-	0791-	1131-	2141-	3171-	3191-
"	47	±5	"	"	"	"	"	70	-28	+13	+15	1155	"	0132-	0352-	0572-	0792-	1132-	2142-	3172-	3192-
"	60	±20	2	12	12	"	"	45	-16	+10.5	+12	1335	T3	0133-	0353-	0573-	0793-	1133-	2143-	3173-	3193-
"	60	±10	"	"	"	"	"	"	-16	+10.5	+12	1335	"	0134-	0354-	0574-	0794-	1134-	2144-	3174-	3194-
"	60	±5	"	"	"	"	"	"	-16	+10.5	+12	1335	"	0135-	0355-	0575-	0795-	1135-	2145-	3175-	3195-
"	82	±20	"	16	15	"	"	"	-32	+13	+15	1400	"	0136-	0356-	0576-	0796-	1136-	2146-	3176-	3196-
"	82	±10	"	"	"	"	"	"	-32	+13	+15	1400	"	0137-	0357-	0577-	0797-	1137-	2147-	3177-	3197-
"	82	±5	"	"	"	"	"	"	-32	+13	+15	1400	"	0138-	0358-	0578-	0798-	1138-	2148-	3178-	3198-
"	160	±20	8	32	17	"	"	27	-50	+25	+25	1900	T4	0139-	0359-	0579-	0799-	1139-	2149-	3179-	3199-
"	160	±10	8	32	17	"	"	27	-50	+25	+25	1900	T4	0140-	0360-	0580-	0800-	1140-	2150-	3180-	3200-
60	4	±20	1	2	2.8	40	69.0	550	-16	+5	+6	525	T1	0141-	0361-	0581-	0801-	1141-	2151-	3181-	3201-
"	4	±10	"	"	"	"	"	550	-16	+5	+6	525	"	0142-	0362-	0582-	0802-	1142-	2152-	3182-	3202-
"	4	±5	"	"	"	"	"	550	-16	+5	+6	525	"	0143-	0363-	0583-	0803-	1143-	2153-	3183-	3203-
"	8.2	±20	"	"	4	"	"	275	-24	+8	+9	625	"	0144-	0364-	0584-	0804-	1144-	2154-	3184-	3204-
"	8.2	±10	"	"	"	"	"	275	-24	+8	+9	625	"	0145-	0365-	0585-	0805-	1145-	2155-	3185-	3205-
"	8.2	±5	"	"	"	"	"	275	-24	+8	+9	625	"	0146-	0366-	0586-	0806-	1146-	2156-	3186-	3206-
"	20	±20	"	5	7	"	"	105	-16	+10.5	+12	930	T2	0147-	0367-	0587-	0807-	1147-	2157-	3187-	3207-
"	20	±10	"	"	"	"	"	105	-16	"	"	930	"	0148-	0368-	0588-	0808-	1148-	2158-	3188-	3208-
"	20	±5	"	"	"	"	"	105	-16	"	"	930	"	0149-	0369-	0589-	0809-	1149-	2159-	3189-	3209-
"	39	±20	"	9	10	"	"	90	-28	"	"	1110	"	0150-	0370-	0590-	0810-	1150-	2160-	3190-	3210-
"	39	±10	"	"	"	"	"	90	-28	"	"	1110	"	0151-	0371-	0591-	0811-	1151-	2161-	3191-	3211-
"	39	±5	"	"	"	"	"	90	-28	"	"	1110	"	0152-	0372-	0592-	0812-	1152-	2162-	3192-	3212-
"	50	±20	2	12	"	"	"	50	-16	"	"	1330	T3	0153-	0373-	0593-	0813-	1153-	2163-	3193-	3213-
"	50	±10	"	"	"	"	"	50	-16	"	"	1330	"	0154-	0374-	0594-	0814-	1154-	2164-	3194-	3214-
"	50	±5	"	"	"	"	"	50	-16	"	"	1330	"	0155-	0375-	0595-	0815-	1155-	2165-	3195-	3215-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.) +25°C	DC leakage (max.) +85°C & +125°C	Dissipation factor (max.)	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at -55°C	Capacitance change at +85°C	Capacitance change at +125°C	2/ Max 85°C 40 kHz ripple current	Case size	Part no. M39006/22- 3/ noncompression seal Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)	Part no. M39006/22- 3/ compression seal Failure rate level (%/1,000 hr.) M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	V dc	V dc	Ohms	%	%	%	mA rms									
60	68	±20	2	16	13	40	69.0	50	-32	+10.5	+12	1365	T3	0156-	0376-	0596-	0816-	1156-	2166-	3196-	3216-
"	68	±10	"	"	"	"	"	50	-32	+10.5	+12	1365	"	0157-	0377-	0597-	0817-	1157-	2167-	3197-	3217-
"	68	±5	"	"	"	"	"	50	-32	+10.5	+12	1365	"	0158-	0378-	0598-	0818-	1158-	2168-	3198-	3218-
"	140	±20	8	32	16	"	"	28	-40	+20	+20	1850	T4	0159-	0379-	0599-	0819-	1159-	2169-	3199-	3219-
"	140	±10	8	32	16	"	"	28	-40	+20	+20	1850	T4	0160-	0380-	0600-	0820-	1160-	2170-	3200-	3220-
75	3.5	±20	1	2	2.5	50	86.2	650	-16	+5	+6	525	T1	0161-	0381-	0601-	0821-	1161-	2171-	3201-	3221-
"	3.5	±10	"	"	2.5	"	"	650	-16	+5	+6	525	"	0162-	0382-	0602-	0822-	1162-	2172-	3202-	3222-
"	3.5	±5	"	"	2.5	"	"	650	-16	+5	+6	525	"	0163-	0383-	0603-	0823-	1163-	2173-	3203-	3223-
"	6.8	±20	"	"	3.5	"	"	300	-20	+8	+9	610	"	0164-	0384-	0604-	0824-	1164-	2174-	3204-	3224-
"	6.8	±10	"	"	3.5	"	"	300	-20	"	"	610	"	0165-	0385-	0605-	0825-	1165-	2175-	3205-	3225-
"	6.8	±5	"	"	3.5	"	"	300	-20	"	"	610	"	0166-	0386-	0606-	0826-	1166-	2176-	3206-	3226-
"	15	±20	"	5	6	"	"	150	-16	"	"	890	T2	0167-	0387-	0607-	0827-	1167-	2177-	3207-	3227-
"	15	±10	"	"	"	"	"	150	-16	"	"	890	"	0168-	0388-	0608-	0828-	1168-	2178-	3208-	3228-
"	15	±5	"	"	"	"	"	150	-16	"	"	890	"	0169-	0389-	0609-	0829-	1169-	2179-	3209-	3229-
"	33	±20	"	10	10	"	"	90	-24	+10.5	+15	1000	"	0170-	0390-	0610-	0830-	1170-	2180-	3210-	3230-
"	33	±10	"	"	"	"	"	90	-24	"	"	1000	"	0171-	0391-	0611-	0831-	1171-	2181-	3211-	3231-
"	33	±5	"	"	"	"	"	90	-24	"	"	1000	"	0172-	0392-	0612-	0832-	1172-	2182-	3212-	3232-
"	40	±20	2	12	9	"	"	60	-16	"	+12	1250	T3	0173-	0393-	0613-	0833-	1173-	2183-	3213-	3233-
"	40	±10	"	"	"	"	"	"	-16	"	"	1250	"	0174-	0394-	0614-	0834-	1174-	2184-	3214-	3234-
"	40	±5	"	"	"	"	"	"	-16	"	"	1250	"	0175-	0395-	0615-	0835-	1175-	2185-	3215-	3235-
"	56	±20	"	17	11	"	"	"	-28	"	+15	1335	"	0176-	0396-	0616-	0836-	1176-	2186-	3216-	3236-
"	56	±10	"	"	"	"	"	"	-28	"	"	1335	"	0177-	0397-	0617-	0837-	1177-	2187-	3217-	3237-
"	56	±5	"	"	"	"	"	"	-28	"	"	1335	"	0178-	0398-	0618-	0838-	1178-	2188-	3218-	3238-
"	110	±20	9	36	12	"	"	29	-35	+20	+20	1850	T4	0179-	0399-	0619-	0839-	1179-	2189-	3219-	3239-
"	110	±10	9	36	12	"	"	29	-35	+20	+20	1850	T4	0180-	0400-	0620-	0840-	1180-	2190-	3220-	3240-
100	2.5	±20	1	2	2	65	115	950	-16	+7	+8	505	T1	0181-	0401-	0621-	0841-	1181-	2191-	3221-	3241-
"	2.5	±10	"	"	"	"	"	950	"	"	"	505	"	0182-	0402-	0622-	0842-	1182-	2192-	3222-	3242-
"	2.5	±5	"	"	"	"	"	950	"	"	"	505	"	0183-	0403-	0623-	0843-	1183-	2193-	3223-	3243-
"	4.7	±20	"	3	"	"	"	500	"	"	"	565	"	0184-	0404-	0624-	0844-	1184-	2194-	3224-	3244-
"	4.7	±10	"	"	"	"	"	500	"	"	"	565	"	0185-	0405-	0625-	0845-	1185-	2195-	3225-	3245-
"	4.7	±5	"	"	"	"	"	500	"	"	"	565	"	0186-	0406-	0626-	0846-	1186-	2196-	3226-	3246-
"	11	±20	"	4	5	"	"	200	"	+8	"	835	T2	0187-	0407-	0627-	0847-	1187-	2197-	3227-	3247-
"	11	±10	"	"	"	"	"	200	"	"	"	835	"	0188-	0408-	0628-	0848-	1188-	2198-	3228-	3248-
"	11	±5	"	"	"	"	"	200	"	"	"	835	"	0189-	0409-	0629-	0849-	1189-	2199-	3229-	3249-
"	22	±20	"	9	7.5	"	"	100	"	"	"	965	"	0190-	0410-	0630-	0850-	1190-	2200-	3230-	3250-
"	22	±10	"	"	"	"	"	100	"	"	"	965	"	0191-	0411-	0631-	0851-	1191-	2201-	3231-	3251-
"	22	±5	"	"	"	"	"	100	"	"	"	965	"	0192-	0412-	0632-	0852-	1192-	2202-	3232-	3252-
"	30	±20	2	12	7	"	"	80	"	"	"	1240	T3	0193-	0413-	0633-	0853-	1193-	2203-	3233-	3253-
"	30	±10	"	"	"	"	"	80	"	"	"	1240	"	0194-	0414-	0634-	0854-	1194-	2204-	3234-	3254-
"	30	±5	"	"	"	"	"	80	"	"	"	1240	"	0195-	0415-	0635-	0855-	1195-	2205-	3235-	3255-
"	43	±20	"	17	8.5	"	"	70	-20	"	"	1335	"	0196-	0416-	0636-	0856-	1196-	2206-	3236-	3256-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max.)		Dissipation factor (max.)	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max.)	Capacitance change at			2/ Max 85°C 40 kHz ripple current	Case size	Part no. M39006/22- 3/ noncompression seal				Part no. M39006/22- 3/ compression seal			
			+25°C	+85°C & +125°C					-55°C	+85°C	+125°C			Failure rate level (%/1,000 hr.)				Failure rate level (%/1,000 hr.)			
														M(1.0)	P(0.1)	R(0.01)	S(0.001)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	V dc	V dc	Ohms	%	%	%	mA rms									
100	43	±10	2	17	8.5	65	115	70	-20	+8	+8	1335	T3	0197-	0417-	0637-	0857-	1197-	2207-	3237-	3257-
"	43	±5	2	17	8.5	"	"	70	-20	+8	+8	1335	T3	0198-	0418-	0638-	0858-	1198-	2208-	3238-	3258-
"	86	±20	9	36	10	"	"	30	-25	+15	+15	1800	T4	0199-	0419-	0639-	0859-	1199-	2209-	3239-	3259-
"	86	±10	9	36	10	"	"	30	-25	+15	+15	1800	T4	0200-	0420-	0640-	0860-	1200-	2210-	3240-	3260-
125	1.7	±20	1	2	2	85	144	1250	-16	+7	+8	415	T1	0201-	0421-	0641-	0861-	1201-	2211-	3241-	3261-
"	1.7	±10	"	"	"	"	"	1250	"	"	"	415	"	0202-	0422-	0642-	0862-	1202-	2212-	3242-	3262-
"	1.7	±5	"	"	"	"	"	1250	"	"	"	415	"	0203-	0423-	0643-	0863-	1203-	2213-	3243-	3263-
"	3.6	±20	"	"	2.7	"	"	600	"	"	"	520	"	0204-	0424-	0644-	0864-	1204-	2214-	3244-	3264-
"	3.6	±10	"	"	2.7	"	"	600	"	"	"	520	"	0205-	0425-	0645-	0865-	1205-	2215-	3245-	3265-
"	3.6	±5	"	"	2.7	"	"	600	"	"	"	520	"	0206-	0426-	0646-	0866-	1206-	2216-	3246-	3266-
"	9	±20	"	5	5	"	"	240	"	"	"	755	T2	0207-	0427-	0647-	0867-	1207-	2217-	3247-	3267-
"	9	±10	"	"	"	"	"	240	"	"	"	755	"	0208-	0428-	0648-	0868-	1208-	2218-	3248-	3268-
"	9	±5	"	"	"	"	"	240	"	"	"	755	"	0209-	0429-	0649-	0869-	1209-	2219-	3249-	3269-
"	14	±20	"	7	6	"	"	167	"	"	"	860	"	0210-	0430-	0650-	0870-	1210-	2220-	3250-	3270-
"	14	±10	"	"	"	"	"	167	"	"	"	860	"	0211-	0431-	0651-	0871-	1211-	2221-	3251-	3271-
"	14	±5	"	"	"	"	"	167	"	"	"	860	"	0212-	0432-	0652-	0872-	1212-	2222-	3252-	3272-
"	18	±20	2	9	5	"	"	129	"	"	"	1130	T3	0213-	0433-	0653-	0873-	1213-	2223-	3253-	3273-
"	18	±10	"	"	"	"	"	129	"	"	"	1130	"	0214-	0434-	0654-	0874-	1214-	2224-	3254-	3274-
"	18	±5	"	"	"	"	"	129	"	"	"	1130	"	0215-	0435-	0655-	0875-	1215-	2225-	3255-	3275-
"	25	±20	"	13	6	"	"	93	"	"	"	1200	"	0216-	0436-	0656-	0876-	1216-	2226-	3256-	3276-
"	25	±10	"	"	"	"	"	93	"	"	"	1200	"	0217-	0437-	0657-	0877-	1217-	2227-	3257-	3277-
"	25	±5	"	"	"	"	"	93	"	"	"	1200	"	0218-	0438-	0658-	0878-	1218-	2228-	3258-	3278-
"	56	±20	10	40	6.5	"	"	32	-25	+15	+15	1800	T4	0219-	0439-	0659-	0879-	1219-	2229-	3259-	3279-
"	56	±10	10	40	6.5	"	"	32	-25	+15	+15	1800	T4	0220-	0440-	0660-	0880-	1220-	2230-	3260-	3280-

1/ Reverse voltage rating at 85°C is 3 V dc, and at 125°C is 2 V dc.

2/ For ripple current limits at various temperatures, voltages, and frequencies, see table II.

3/ Dash number will include the letter "H" to indicate the optional vibration and shock requirements (i.e., 51 g's random vibration, 80 g's sinusoidal vibration, and 500 g's shock) or "-" will be deleted.

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TABLE II. Ripple current multipliers versus frequency, temperature, and applied peak voltage. 1/ 2/ 3/ 4/ 5/

Frequency of applied ripple current	120 Hz				800 Hz				1 kHz				10 kHz				40 kHz				100 kHz			
	Temp °C				Temp °C				Temp °C				Temp °C				Temp °C				Temp °C			
Ambient still air	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°
100%	.60	.39	---	---	.71	.43	---	---	.72	.45	---	---	.88	.55	---	---	1.0	.63	---	---	1.1	.69	---	---
90%	.60	.46	---	---	.71	.55	---	---	.72	.55	---	---	.88	.67	---	---	1.0	.77	---	---	1.1	.85	---	---
80%	.60	.52	.35	---	.71	.62	.42	---	.72	.62	.42	---	.88	.76	.52	---	1.0	.87	.59	---	1.1	.96	.65	---
70%	.60	.58	.44	---	.71	.69	.52	---	.72	.70	.52	---	.88	.85	.64	---	1.0	.97	.73	---	1.1	1.07	.80	---
≤66 2/3%	.60	.60	.46	.27	.71	.71	.55	.32	.72	.72	.55	.32	.88	.88	.68	.40	1.0	1.0	.77	.45	1.1	1.1	.85	.50

1/ At 125°C the rated voltage of the capacitors decreases to 66 2/3 percent of the 85°C rated voltage.
2/ The peak of the applied ac ripple voltage plus the applied dc voltage must not exceed the dc voltage rating of the capacitor either forward or reverse.
3/ The ripple current listed in table I represents a rating calculated using a maximum internal temperature rise (ΔT) of 50°C at 40 kHz at 85°C ambient with a maximum peak rated voltage of 66 2/3 percent of the 85°C peak voltage rating.
4/ The maximum allowable internal temperature rise (ΔT) decreases linearly to a calculated 10°C rise at 125°C ambient.
5/ The internal temperature rise is directly proportional to the ESR of the capacitor, and ESR increases with decreasing frequency.

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TABLE III. Substitutability data. 1/

Dash number in MIL-C-39006/22B, /22C, and /22D				Substitutable for dash no. MIL-C-39006/22A and /22					Dash number in MIL-C-39006/22B and /22C				Substitutable for dash no. in MIL-C-39006/22A and /22				
M	P	R	S	L 2/	M	P	R	S	M	P	R	S	L 2/	M	P	R	S
0001	0221	0441	0661	6001	6206	6411	6616	6821	0033	0253	0473	0693	---	---	---	---	---
0002	0222	0442	0662	6002	6207	6412	6617	6822	0034	0254	0474	0694	---	---	---	---	---
0003	0223	0443	0663	6003	6208	6413	6618	6823	0035	0255	0475	0695	---	---	---	---	---
0004	0224	0444	0664	6004	6209	6414	6619	6824	0036	0256	0476	0696	6030	6235	6440	6645	6850
0005	0225	0445	0665	6005	6210	6415	6620	6825	0037	0257	0477	0697	6031	6236	6441	6646	6851
0006	0226	0446	0666	6006	6211	6416	6621	6826	0038	0258	0478	0698	6032	6237	6442	6647	6852
0007	0227	0447	0667	6007	6212	6417	6622	6827	---	---	---	---	7049	7131	7213	7295	7377
0008	0228	0448	0668	6008	6213	6418	6623	6828	---	---	---	---	7050	7132	7214	7296	7378
0009	0229	0449	0669	6009	6214	6419	6624	6829	---	---	---	---	7051	7133	7215	7297	7379
0010	0230	0450	0670	6010	6215	6420	6625	6830	0039	0259	0479	0699	6033	6238	6443	6648	6853
0011	0231	0451	0671	6011	6216	6421	6626	6831	0040	0260	0480	0700	6034	6239	6444	6649	6854
0012	0232	0452	0672	6012	6217	6422	6627	6832	---	---	---	---	7052	7134	7216	7298	7380
0013	0233	0453	0673	6013	6218	6423	6628	6833	---	---	---	---	7053	7135	7217	7299	7381
0014	0234	0454	0674	6014	6219	6424	6629	6834	0041	0261	0481	0701	6035	6240	6445	6650	6855
0015	0235	0455	0675	6015	6220	6425	6630	6835	0042	0262	0482	0702	6036	6241	6446	6651	6856
---	---	---	---	7041	7123	7205	7287	7369	0043	0263	0483	0703	6037	6242	6447	6652	6857
---	---	---	---	7042	7124	7206	7288	7370	0044	0264	0484	0704	6038	6243	6448	6653	6858
---	---	---	---	7043	7125	7207	7289	7371	0045	0265	0485	0705	6039	6244	6449	6654	6859
0016	0236	0456	0676	6016	6221	6426	6631	6836	0046	0266	0486	0706	6040	6245	6450	6655	6860
0017	0237	0457	0677	6017	6222	6427	6632	6837	0047	0267	0487	0707	6041	6246	6451	6656	6861
0018	0238	0458	0678	6018	6223	6428	6633	6838	0048	0268	0488	0708	6042	6247	6452	6657	6862
---	---	---	---	7044	7126	7208	7290	7372	0049	0269	0489	0709	6043	6248	6453	6658	6863
---	---	---	---	7045	7127	7209	7291	7373	0050	0270	0490	0710	6044	6249	6454	6659	6864
---	---	---	---	7046	7128	7210	7292	7374	0051	0271	0491	0711	6045	6250	6455	6660	6865
0019	0239	0459	0679	6019	6224	6429	6634	6839	0052	0272	0492	0712	6046	6251	6456	6661	6866
0020	0240	0460	0680	6020	6225	6430	6635	6840	0053	0273	0493	0713	6047	6252	6457	6662	6867
---	---	---	---	7047	7129	7211	7293	7375	0054	0274	0494	0714	6048	6253	6458	6663	6868
---	---	---	---	7048	7130	7212	7294	7376	0055	0275	0495	0715	6049	6254	6459	6664	6869
0021	0241	0461	0681	6021	6226	6431	6636	6841	---	---	---	---	7054	7136	7218	7300	7382
0022	0242	0462	0682	6022	6227	6432	6637	6842	---	---	---	---	7055	7137	7219	7301	7383
0023	0243	0463	0683	6023	6228	6433	6638	6843	---	---	---	---	7056	7138	7220	7302	7384
0024	0244	0464	0684	6024	6229	6434	6639	6844	0056	0276	0496	0716	6050	6255	6460	6665	6870
0025	0245	0465	0685	6025	6230	6435	6640	6845	0057	0277	0497	0717	6051	6256	6461	6666	6871
0026	0246	0466	0686	6026	6231	6436	6641	6846	0058	0278	0498	0718	6052	6257	6462	6667	6872
0027	0247	0467	0687	---	---	---	---	---	---	---	---	---	7057	7139	7221	7303	7385
0028	0248	0468	0688	---	---	---	---	---	---	---	---	---	7058	7140	7222	7304	7386
0029	0249	0469	0689	---	---	---	---	---	---	---	---	---	7059	7141	7223	7305	7387
0030	0250	0470	0690	6027	6232	6437	6642	6847	0059	0279	0499	0719	6053	6258	6463	6668	6873
0031	0251	0471	0691	6028	6233	6438	6643	6848	0060	0280	0500	0720	6054	6259	6464	6669	6874
0032	0252	0472	0692	6029	6234	6439	6644	6849	---	---	---	---	7060	7142	7224	7306	7388
---	---	---	---	7451	7457	7463	7469	7475	---	---	---	---	7061	7143	7225	7307	7389
---	---	---	---	7452	7458	7464	7470	7476	0061	0281	0501	0721	6055	6260	6465	6670	6875
---	---	---	---	7453	7459	7465	7471	7477									

See footnotes at end of table.

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TABLE III. Substitutability data - Continued. 1/

Dash number in MIL-C-39006/22B, /22C, and /22D				Substitutable for dash no. MIL-C-39006/22A and /22					Dash number in MIL-C-39006/22B and /22C				Substitutable for dash no. in MIL-C-39006/22A and /22				
M	P	R	S	L 2/	M	P	R	S	M	P	R	S	L 2/	M	P	R	S
0062	0282	0502	0722	6056	6261	6466	6671	6876	0094	0314	0534	0754	---	---	---	---	---
0063	0283	0503	0723	6057	6262	6467	6672	6877	0095	0315	0535	0755	---	---	---	---	---
0064	0284	0504	0724	6058	6263	6468	6673	6878	0096	0316	0536	0756	6084	6289	6494	6699	6904
0065	0285	0505	0725	6059	6264	6469	6674	6879	0097	0317	0537	0757	6085	6290	6495	6700	6905
0066	0286	0506	0726	6060	6265	6470	6675	6880	0098	0318	0538	0758	6086	6291	6496	6701	6906
0067	0287	0507	0727	6061	6266	6471	6676	6881	---	---	---	---	7070	7152	7234	7316	7398
0068	0288	0508	0728	6062	6267	6472	6677	6882	---	---	---	---	7071	7153	7235	7317	7399
0069	0289	0509	0729	6063	6268	6473	6678	6883	---	---	---	---	7072	7154	7236	7318	7400
0070	0290	0510	0730	6064	6269	6474	6679	6884	0099	0319	0539	0759	6087	6292	6497	6702	6907
0071	0291	0511	0731	6065	6270	6475	6680	6885	0100	0320	0540	0760	6088	6293	6498	6703	6908
0072	0292	0512	0732	6066	6271	6476	6681	6886	---	---	---	---	7073	7155	7237	7319	7401
0073	0293	0513	0733	6067	6272	6477	6682	6887	---	---	---	---	7074	7156	7238	7320	7402
0074	0294	0514	0734	6068	6273	6478	6683	6888	0101	0321	0541	0761	6089	6294	6499	6704	6909
0075	0295	0515	0735	6069	6274	6479	6684	6889	0102	0322	0542	0762	6090	6295	6500	6705	6910
---	---	---	---	7062	7144	7226	7308	7390	0103	0323	0543	0763	6091	6296	6501	6706	6911
---	---	---	---	7063	7145	7227	7309	7391	0104	0324	0544	0764	6092	6297	6502	6707	6912
---	---	---	---	7064	7146	7228	7310	7392	0105	0325	0545	0765	6093	6298	6503	6708	6913
0076	0296	0516	0736	6070	6275	6480	6685	6890	0106	0326	0546	0766	6094	6299	6504	6709	6914
0077	0297	0517	0737	6071	6276	6481	6686	6891	0107	0327	0547	0767	6095	6300	6505	6710	6915
0078	0298	0518	0738	6072	6277	6482	6687	6892	0108	0328	0548	0768	6096	6301	6506	6711	6916
---	---	---	---	7065	7147	7229	7311	7393	0109	0329	0549	0769	6097	6302	6507	6712	6917
---	---	---	---	7066	7148	7230	7312	7394	0110	0330	0550	0770	6098	6303	6508	6713	6918
---	---	---	---	7067	7149	7231	7313	7395	0111	0331	0551	0771	6099	6304	6509	6714	6919
0079	0299	0519	0739	6073	6278	6483	6688	6893	0112	0332	0552	0772	6100	6305	6510	6715	6920
0080	0300	0520	0740	6074	6279	6484	6689	6894	0113	0333	0553	0773	6101	6306	6511	6716	6921
---	---	---	---	7068	7150	7232	7314	7396	0114	0334	0554	0774	6102	6307	6512	6717	6922
---	---	---	---	7069	7151	7233	7315	7397	0115	0335	0555	0775	6103	6308	6513	6718	6923
0081	0301	0521	0741	6075	6280	6485	6690	6895	---	---	---	---	7075	7157	7239	7321	7403
0082	0302	0522	0742	6076	6281	6486	6691	6896	---	---	---	---	7076	7158	7240	7322	7404
0083	0303	0523	0743	6077	6282	6487	6692	6897	---	---	---	---	7077	7159	7241	7323	7405
0084	0304	0524	0744	6078	6283	6488	6693	6898	0116	0336	0556	0776	6104	6309	6514	6719	6924
0085	0305	0525	0745	6079	6284	6489	6694	6899	0117	0337	0557	0777	6105	6310	6515	6720	6925
0086	0306	0526	0746	6080	6285	6490	6695	6900	0118	0338	0558	0778	6106	6311	6516	6721	6926
0087	0307	0527	0747	---	---	---	---	---	---	---	---	---	7078	7160	7242	7324	7406
0088	0308	0528	0748	---	---	---	---	---	---	---	---	---	7079	7161	7243	7325	7407
0089	0309	0529	0749	---	---	---	---	---	---	---	---	---	7080	7162	7244	7326	7408
0090	0310	0530	0750	6081	6286	6491	6696	6901	0119	0339	0559	0779	6107	6312	6517	6722	6927
0091	0311	0531	0751	6082	6287	6492	6697	6902	0120	0340	0560	0780	6108	6313	6518	6723	6928
0092	0312	0532	0752	6083	6288	6493	6698	6903	---	---	---	---	7081	7163	7245	7327	7409
---	---	---	---	7454	7460	7466	7472	7478	---	---	---	---	7082	7164	7246	7328	7410
---	---	---	---	7455	7461	7467	7473	7479	0121	0341	0561	0781	6109	6314	6519	6724	6929
---	---	---	---	7456	7462	7468	7474	7480	0122	0342	0562	0782	6110	6315	6520	6725	6930
0093	0313	0533	0753	---	---	---	---	---	0123	0343	0563	0783	6111	6316	6521	6726	6931

See footnotes at end of table.

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TABLE III. Substitutability data - Continued. 1/

Dash number in MIL-C-39006/22B, /22C, and /22D				Substitutable for dash no. MIL-C-39006/22A and /22					Dash number in MIL-C-39006/22B and /22C				Substitutable for dash no. in MIL-C-39006/22A and /22				
M	P	R	S	L 2/	M	P	R	S	M	P	R	S	L 2/	M	P	R	S
0124	0344	0564	0784	6112	6317	6522	6727	6932	0157	0377	0597	0817	6145	6350	6555	6760	6965
0125	0345	0565	0785	6113	6318	6523	6728	6933	0158	0378	0598	0818	6146	6351	6556	6761	6966
0126	0346	0566	0786	6114	6319	6524	6729	6934	---	---	---	---	7094	7176	7258	7340	7422
0127	0347	0567	0787	6115	6320	6525	6730	6935	---	---	---	---	7095	7177	7259	7341	7423
0128	0348	0568	0788	6116	6321	6526	6731	6936	---	---	---	---	7096	7178	7260	7342	7424
0129	0349	0569	0789	6117	6322	6527	6732	6937	0159	0379	0599	0819	6147	6352	6557	6762	6967
0130	0350	0570	0790	6118	6323	6528	6733	6938	0160	0380	0600	0820	6148	6353	6558	6763	6968
0131	0351	0571	0791	6119	6324	6529	6734	6939	---	---	---	---	7097	7179	7261	7343	7425
0132	0352	0572	0792	6120	6325	6530	6735	6940	---	---	---	---	7098	7180	7261	7344	7426
0133	0353	0573	0793	6121	6326	6531	6736	6941	0161	0381	0601	0821	6149	6354	6559	6764	6969
0134	0354	0574	0794	6122	6327	6532	6737	6942	0162	0382	0602	0822	6150	6355	6560	6765	6970
0135	0355	0575	0795	6123	6328	6533	6738	6943	0163	0383	0603	0823	6151	6356	6561	6766	6971
---	---	---	---	7083	7165	7247	7329	7411	0164	0384	0604	0824	6152	6357	6562	6767	6972
---	---	---	---	7084	7166	7248	7330	7412	0165	0385	0605	0825	6153	6358	6563	6768	6973
---	---	---	---	7085	7167	7249	7331	7413	0166	0386	0606	0826	6154	6359	6564	6769	6974
0136	0356	0576	0796	6124	6329	6534	6739	6944	0167	0387	0607	0827	6155	6360	6565	6770	6975
0137	0357	0577	0797	6125	6330	6535	6740	6945	0168	0388	0608	0828	6156	6361	6566	6771	6976
0138	0358	0578	0798	6126	6331	6536	6741	6946	0169	0389	0609	0829	6157	6362	6567	6772	6977
---	---	---	---	7086	7168	7250	7332	7414	0170	0390	0610	0830	6158	6363	6568	6773	6978
---	---	---	---	7087	7169	7251	7333	7415	0171	0391	0611	0831	6159	6364	6569	6774	6979
---	---	---	---	7088	7170	7252	7334	7416	0172	0392	0612	0832	6160	6365	6570	6775	6980
0139	0359	0579	0799	6127	6332	6537	6742	6947	0173	0393	0613	0833	6161	6366	6571	6776	6981
0140	0360	0580	0800	6128	6333	6538	6743	6948	0174	0394	0614	0834	6162	6367	6572	6777	6982
---	---	---	---	7089	7171	7253	7335	7417	0175	0395	0615	0835	6163	6368	6573	6778	6983
---	---	---	---	7090	7172	7254	7336	7418	---	---	---	---	7099	7181	7263	7345	7427
0141	0361	0581	0801	6129	6334	6539	6744	6949	---	---	---	---	7100	7182	7264	7346	7428
0142	0362	0582	0802	6130	6335	6540	6745	6950	---	---	---	---	7101	7183	7265	7347	7429
0143	0363	0583	0803	6131	6336	6541	6746	6951	0176	0396	0616	0836	6164	6369	6574	6779	6984
0144	0364	0584	0804	6132	6337	6542	6747	6952	0177	0397	0617	0837	6165	6370	6575	6780	6985
0145	0365	0585	0805	6133	6338	6543	6748	6953	0178	0398	0618	0838	6166	6371	6576	6781	6986
0146	0366	0586	0806	6134	6339	6544	6749	6954	---	---	---	---	7102	7184	7266	7348	7430
0147	0367	0587	0807	6135	6340	6545	6750	6955	---	---	---	---	7103	7185	7267	7349	7431
0148	0368	0588	0808	6136	6341	6546	6751	6956	---	---	---	---	7104	7186	7268	7350	7432
0149	0369	0589	0809	6137	6342	6547	6752	6957	0179	0399	0619	0839	6167	6372	6577	6782	6987
0150	0370	0590	0810	6138	6343	6548	6753	6958	0180	0400	0620	0840	6168	6373	6578	6783	6988
0151	0371	0591	0811	6139	6344	6549	6754	6959	---	---	---	---	7105	7187	7269	7351	7433
0152	0372	0592	0812	6140	6345	6550	6755	6960	---	---	---	---	7106	7188	7270	7352	7434
0153	0373	0593	0813	6141	6346	6551	6756	6961	0181	0401	0621	0841	6169	6374	6579	6784	6989
0154	0374	0594	0814	6152	6347	6552	6757	6962	0182	0402	0622	0842	6170	6375	6580	6785	6990
0155	0375	0595	0815	6153	6348	6553	6758	6963	0183	0403	0623	0843	6171	6376	6581	6786	6991
---	---	---	---	7091	7173	7255	7337	7419	0184	0404	0624	0844	6172	6377	6582	6787	6992
---	---	---	---	7092	7174	7256	7338	7420	0185	0405	0625	0845	6173	6378	6583	6788	6993
---	---	---	---	7093	7175	7257	7339	7421	0186	0406	0626	0846	6174	6379	6584	6789	6994
0156	0376	0596	0816	6144	6349	6554	6759	6964	0187	0407	0627	0847	6145	6380	6585	6790	6995

See footnotes at end of table.

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TABLE III. Substitutability data - Continued. 1/

Dash number in MIL-C-39006/22B, /22C, and /22D				Substitutable for dash no. MIL-C-39006/22A and /22					Dash number in MIL-C-39006/22B and /22C				Substitutable for dash no. in MIL-C-39006/22A and /22				
M	P	R	S	L 2/	M	P	R	S	M	P	R	S	L 2/	M	P	R	S
0188	0408	0628	0848	6176	6381	6586	6791	6996	0205	0425	0645	0865	6193	6398	6603	6808	7013
0189	0409	0629	0849	6177	6382	6587	6792	6997	0206	0426	0646	0866	6194	6399	6604	6809	7014
0190	0410	0630	0850	6178	6383	6588	6793	6998	0207	0427	0647	0867	6195	6400	6605	6810	7015
0191	0411	0631	0851	6179	6384	6589	6794	6999	0208	0428	0648	0868	6196	6401	6606	6811	7016
0192	0412	0632	0852	6180	6385	6590	6795	7000	0209	0429	0649	0869	6197	6402	6607	6812	7017
0193	0413	0633	0853	6181	6386	6591	6796	7001	0210	0430	0650	0870	6198	6403	6608	6813	7018
0194	0414	0634	0854	6182	6387	6592	6797	7002	0211	0431	0651	0871	6199	6404	6609	6814	7019
0195	0415	0635	0855	6183	6388	6593	6798	7003	0212	0432	0652	0872	6200	6405	6610	6815	7020
---	---	---	---	7107	7189	7271	7353	7435	0213	0433	0653	0873	6201	6406	6611	6816	7021
---	---	---	---	7108	7190	7272	7354	7436	0214	0434	0654	0874	6202	6407	6612	6817	7022
---	---	---	---	7109	7191	7273	7355	7437	0215	0435	0655	0875	6203	6408	6613	6818	7023
0196	0416	0636	0856	6184	6389	6594	6799	7004	---	---	---	---	7115	7197	7279	7361	7443
0197	0417	0637	0857	6185	6390	6595	6800	7005	---	---	---	---	7116	7198	7280	7362	7444
0198	0418	0638	0858	6186	6391	6596	6801	7006	---	---	---	---	7117	7199	7281	7363	7445
---	---	---	---	7110	7192	7274	7356	7438	0216	0436	0656	0876	6204	6409	6614	6819	7024
---	---	---	---	7111	7293	7275	7357	7439	0217	0437	0657	0877	6205	6410	6615	6820	7025
---	---	---	---	7112	7294	7276	7358	7440	0218	0438	0658	0878	6206	6411	6616	6821	7026
0199	0419	0639	0859	6187	6392	6597	6802	7007	---	---	---	---	7118	7200	7282	7364	7446
0200	0420	0640	0860	6188	6393	6598	6803	7008	---	---	---	---	7119	7201	7283	7365	7447
---	---	---	---	7113	7195	7277	7359	7441	---	---	---	---	7120	7202	7284	7366	7448
---	---	---	---	7114	7196	7278	7360	7442	0219	0439	0659	0879	7027	7030	7033	7036	7039
0203	0423	0643	0863	6191	6396	6601	6806	7011	---	---	---	---	7122	7204	7286	7368	7450
0204	0424	0644	0864	6192	6397	6602	6807	7012									

1/ TABLE III, substitutability data are unilateral; i.e., the dash number in MIL-C-39006/22B, /22C, and /22D is substitutable for the dash number in MIL-C-39006/22 and /22A and not vice versa.
2/ Failure rate level "L" is inactive for design, use failure rate level "M" for replacement.

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Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Air Force - 17, 99
DLA - ES

User activities:
Navy - AS, MC, OS
Air Force - 11

Preparing activity:
Air Force - 85

Agent.
DLA - ES

(Project 5910-1545)

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